Week 9 & 10

1. Convex Set

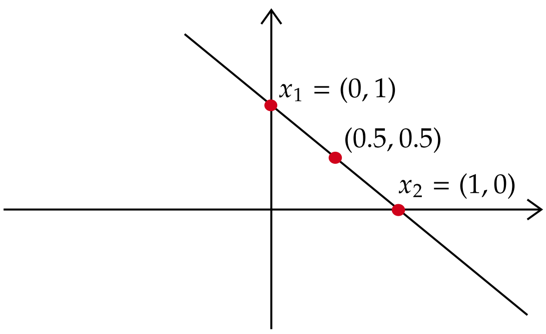
A set $\mathbf{S} \subseteq \mathbb{R}^{\mathbf{d}}$ is a convex set, if, for all $\mathbf{x_1}, \mathbf{x_2} \in \mathbf{S}$, and for all $\lambda \in [0, 1]$

$$(\lambda)x_1 + (1 - \lambda)x_2 \in S$$

NOTE : From here, assume $\lambda \in [0, 1]$, wherever used.

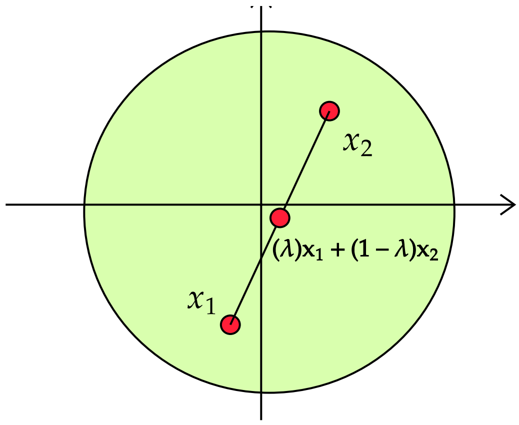
Example

$$S = \{(x, y) | x + y = 1\}$$



Example

$$S = \left\{ x \in \mathbb{R}^2 \middle| ||x||^2 \leqslant 9 \right\}$$



PROOF

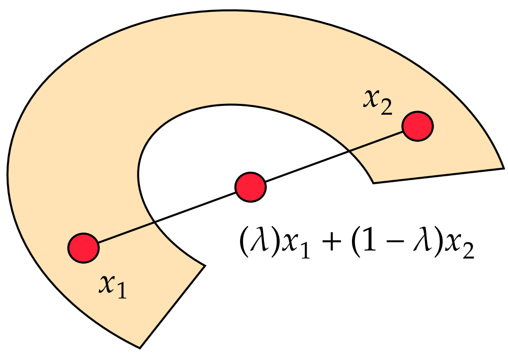
- **Triangular Inequality** : $||u + v|| \leqslant ||u|| + ||v||$, where $u, v \in \mathbb{R}^d$
- $||c \times u|| = |c| \times ||u||$, where $c \in \mathbb{R}$ and $u \in \mathbb{R}^d$

$$\begin{aligned}
||(\lambda)x_1 + (1-\lambda)x_2|| &\leqslant ||(\lambda)x_1|| + ||(1-\lambda)x_2|| \\
&\leqslant (|\lambda|)||x_1|| + (|1-\lambda|)||x_2|| \\
&\leqslant (\lambda)3 + (1-\lambda)3 \\
&\leqslant 3 \\
||(\lambda)x_1 + (1-\lambda)x_2||^2 &\leqslant 9
\end{aligned}$$

Therefore, S is a convex set.

Non-Example

The following is not a convex set



Example

Show that $S = \left\{x \in \mathbb{R}^d \middle| w^T x = b\right\}$ is a convex set.

- Let $x_1, x_2 \in S$, i.e., $w^T x_1 = b$ and $w^T x_2 = b$
- Let $v = (\lambda)x_1 + (1-\lambda)x_2$, $\lambda \in [0,1]$

$$\begin{aligned} w^T v &= w^T[(\lambda)x_1 + (1-\lambda)x_2] \\ &= (\lambda)w^T x_1 + (1-\lambda)w^T x_2 \\ &= (\lambda)b + (1-\lambda)b \\ &= b \end{aligned}$$

$v \in S$. Therefore, S is a convex set.

Example

Show that $S = \left\{x \in \mathbb{R}^d \middle| w^T x \leqslant b\right\}$ is a convex set.

- Let $x_1, x_2 \in S$, i.e., $w^T x_1 < b$ and $w^T x_2 < b$
- Let $v = (\lambda)x_1 + (1-\lambda)x_2$, $\lambda \in [0,1]$

Notice,

- $(\lambda)w^T x_1 \leqslant (\lambda)b$
- $(1-\lambda)w^T x_2 \leqslant (1-\lambda)b$

$$\begin{aligned} w^T v &= w^T[(\lambda)x_1 + (1-\lambda)x_2] \\ &= (\lambda)w^T x_1 + (1-\lambda)w^T x_2 \\ &\leqslant (\lambda)b + (1-\lambda)b \\ &\leqslant b \end{aligned}$$

$v \in S$. Therefore, S is a convex set.

Example

Show that $S = \left\{x \in \mathbb{R}^d \middle| (A_{n\times d})x = b\right\}$ is a convex set.

- Let $x_1, x_2 \in S$, i.e., $(A_{n\times d})x_1 = b$ and $(A_{n\times d})x_2 = b$
- Let $v = (\lambda)x_1 + (1-\lambda)x_2$, $\lambda \in [0,1]$

$$
\begin{aligned}
(A_{n\times d})v &= (A_{n\times d})[(\lambda)x_1 + (1-\lambda)x_2] \\
&= (\lambda)(A_{n\times d})x_1 + (1-\lambda)(A_{n\times d})x_2 \\
&= (\lambda)b + (1-\lambda)b \\
&= b
\end{aligned}
$$

$v \in S$. Therefore, S is a convex set.

A Property Of Convex Set

Intersection of convex sets is convex.

PROOF

- Let S_1 and S_2 be convex sets.
- $x_1, x_2 \in S_1$
- $x_1, x_2 \in S_2$
- Let $x_1, x_2 \in S_1 \cap S_2$

Since S_1 is a convex set, we can say that $(\lambda)x_1 + (1-\lambda)x_2 \in S_1, \lambda \in [0,1]$
Since S_2 is a convex set, we can say that $(\lambda)x_1 + (1-\lambda)x_2 \in S_2, \lambda \in [0,1]$
Therefore, $(\lambda)x_1 + (1-\lambda)x_2 \in S_1 \cap S_2$.

$S_1 \cap S_2$ is a convex set

EXAMPLE

Show that $S = \left\{x \in \mathbb{R}^d \middle| A_{n\times d}x = b\right\}$ is a convex set.

$$
A = \begin{bmatrix} — & a_1 & — \\ — & a_2 & — \\ & \vdots & \\ — & a_n & — \end{bmatrix},\ x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix},\ b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}
$$

$$
\begin{aligned}
S &= \left\{x \in \mathbb{R}^d \middle| A_{n\times d}x = b\right\} \\
&= \left\{x \in \mathbb{R}^d \middle| a_1^T x = b_1, \ldots, a_n^T x = b_n\right\} \\
&= \left\{x \in \mathbb{R}^d | a_1^T x = b_1\right\} \cap \ldots \cap \left\{x \in \mathbb{R}^d | a_n^T x = b_n\right\}
\end{aligned}
$$

$\left\{x \in \mathbb{R}^d | a_i^T x = b_i\right\}$ is a convex set. S is an intersection of convex sets and therefore, S is a convex set.

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2. Convex Combination

Let $S = \{x_1, \ldots, x_n\} \in \mathbb{R}^d$.

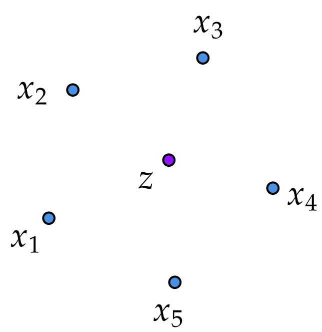
$\mathbf{z} \in \mathbb{R}^d$ is said to be a **convex combination of** $x_1, \ldots, x_n$ if

$$z = \sum_{i=1}^{n} \alpha_i x_i$$

where $\alpha_i \geqslant 0, \ \forall i \in \{1, \ldots, n\}$ and

$$\sum_{i=1}^{n} \alpha_i = 1$$

Example : Let $S = \{x_1, \ldots, x_5\}$.



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3. Convex Hull

Let $S = \{x_1, \ldots, x_n\} \subseteq \mathbb{R}^d$. Then

$$\mathbf{Convex-Hull}(S) = \left\{ z \in \mathbb{R}^d \middle| z = \sum_{i=1}^{n} \alpha_i x_i, \ \alpha_1, \ldots, \alpha_n \geqslant 0, \sum_{i=1}^{n} \alpha_i = 1 \right\}$$

In simple words, it is the set of all possible **convex combinations** of the elements in set S.

$\mathbf{Convex-Hull}(S)$ is a convex set. Proof is given below.

- Let $z_1, z_2 \in \textbf{Convex} - \textbf{Hull}(S)$
- $z_1 = \sum_{i=1}^{n} \alpha_{(1,i)} x_i$ where, $a_{(1,i)} > 0, \ \forall i \in \{1, \ldots, n\}$ and $\sum_{i=1}^{n} \alpha_{1,i} = 1$
- $z_2 = \sum_{i=1}^{n} \alpha_{(2,i)} x_i$ where, $a_{(2,i)} > 0, \ \forall i \in \{1, \ldots, n\}$ and $\sum_{i=1}^{n} \alpha_{2,i} = 1$
- Let z' be $(\lambda) z_1 + (1 - \lambda) z_2$, $\lambda \in [0, 1]$

$$
\begin{aligned}
z' &= (\lambda) z_1 + (1 - \lambda) z_2 \\
&= (\lambda) \sum_{i=1}^{n} \alpha_{(1,i)} x_i + (1 - \lambda) \sum_{i=1}^{n} \alpha_{(2,i)} x_i \\
&= \sum_{i=1}^{n} (\lambda \alpha_{(1,i)}) x_i + \sum_{i=1}^{n} ((1 - \lambda) \alpha_{(2,i)}) x_i \\
&= \sum_{i=1}^{n} \left[(\lambda \alpha_{1,i}) + ((1 - \lambda) \alpha_{2,i}) \right] x_i
\end{aligned}
$$

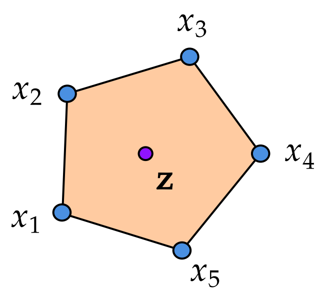
Notice, $(\lambda \alpha_{(1,i)}) + ((1 - \lambda) \alpha_{(2,i)}) \geqslant 0 \ \forall i \in \{1, \ldots, n\}$.

$$
\begin{aligned}
\sum_{i=1}^{n} \left[(\lambda \alpha_{(1,i)}) + ((1 - \lambda) \alpha_{(2,i)}) \right] &= \lambda \sum_{i=1}^{n} \alpha_{(1,i)} + (1 - \lambda) \sum_{i=1}^{n} \alpha_{(2,i)} \\
&= \lambda (1) + (1 - \lambda)(1) \\
&= 1
\end{aligned}
$$

z' is a convex combination of the elements in S. Therefore, $\textbf{Convex} - \textbf{Hull}(S)$ is a convex set.

Example

Let $S = \{x_1, \ldots, x_5\}$



In the above picture, z is a convex combination of $x_1, \ldots, x_5$ and the brown shaded region is the convex hull of S.

ALTERNATE DEFINITION

$\mathbf{Convex-Hull}(\{x_1,..., x_n\})$ is the intersection of all the **convex** sets containing $\{x_1,..., x_n\}$.

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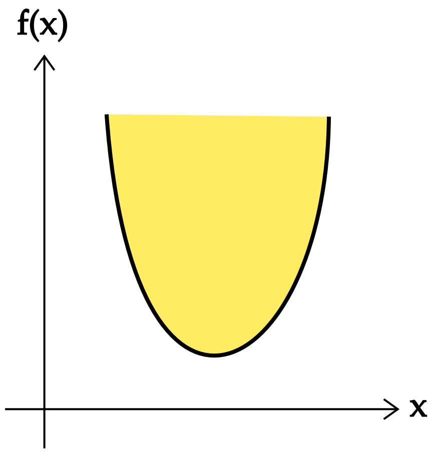
4. Convex Function

DEFINITION 1

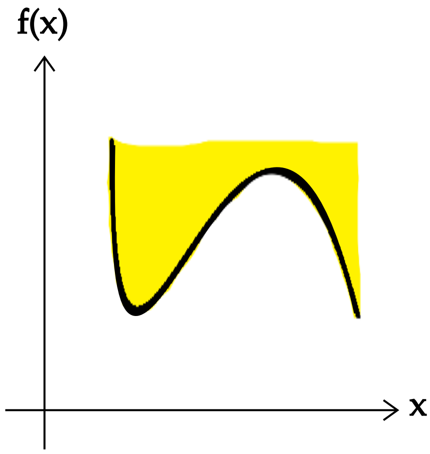
Let $D \subseteq \mathbb{R}^d$ be a convex set. A function $f : D \to \mathbb{R}$ is a convex function if $\mathbf{epi}(f)$ is a convex set.

$$\mathbf{epi}(f) = \left\{ \begin{bmatrix} x \\ z \end{bmatrix} \in \mathbb{R}^{d+1} \middle| z \geqslant f(x) \right\}$$

Here, $\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \cdots & x_d & z \end{bmatrix}^T$.



CONVEX SET

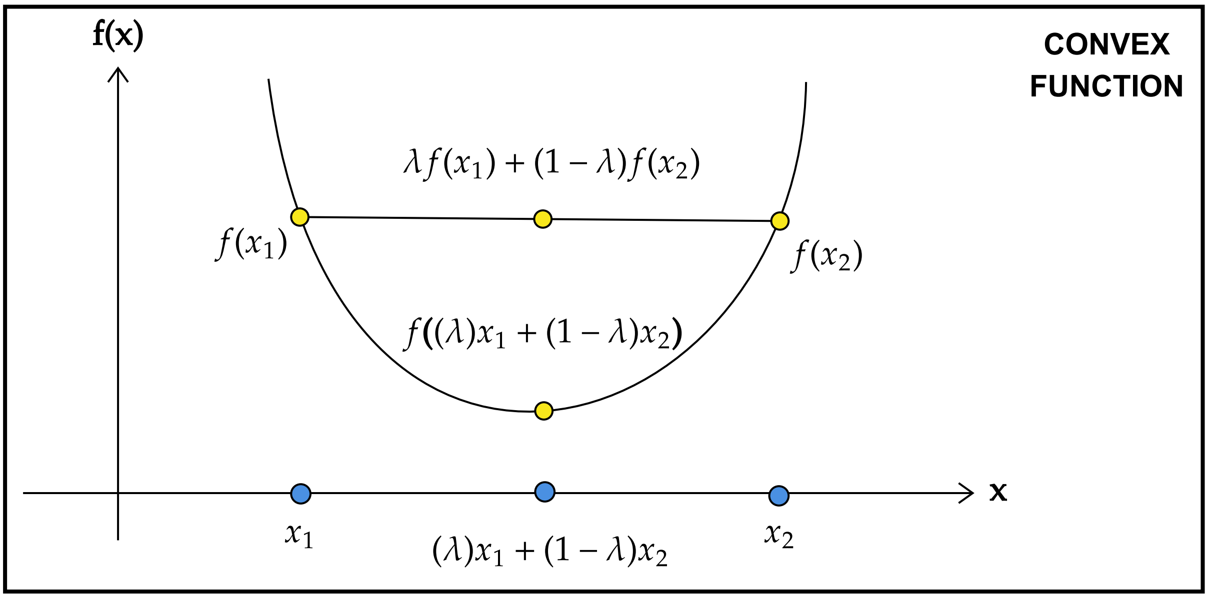


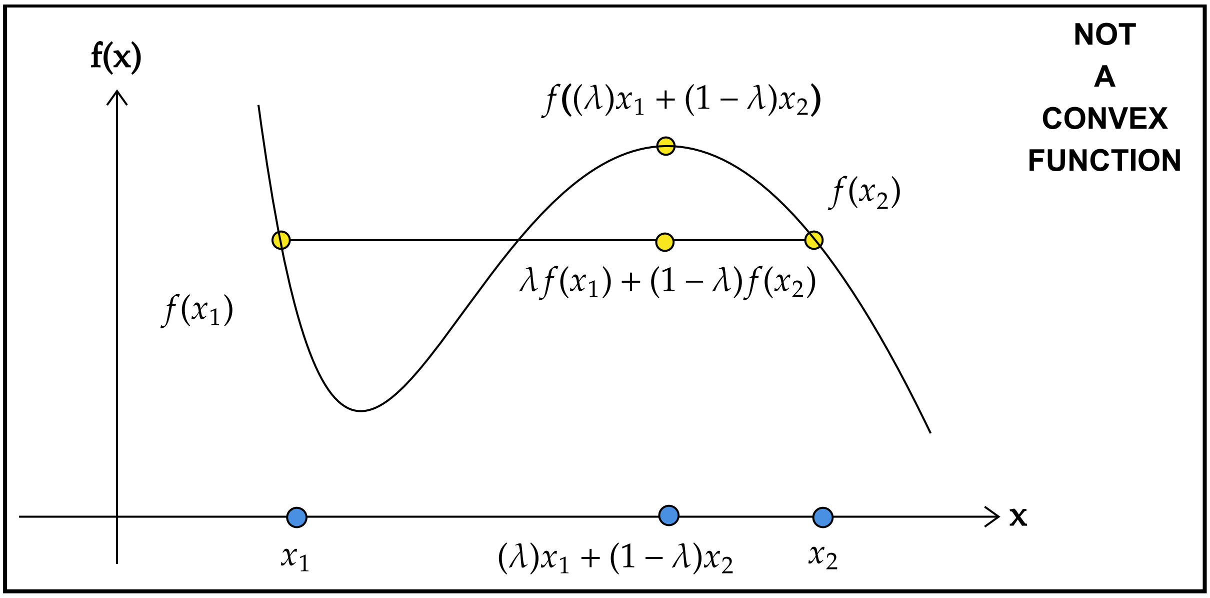
NOT A CONVEX SET

DEFINITION 2

Let $D \subseteq \mathbb{R}^d$ be a convex set. A function $f : D \to \mathbb{R}$ is a convex function if and only if $\forall x_1, x_2 \in \mathbb{R}^d$ and for all $\lambda \in [0, 1]$,

$$f((\lambda)x_1 + (1-\lambda)x_2) \leqslant \lambda f(x_1) + (1-\lambda)f(x_2)$$





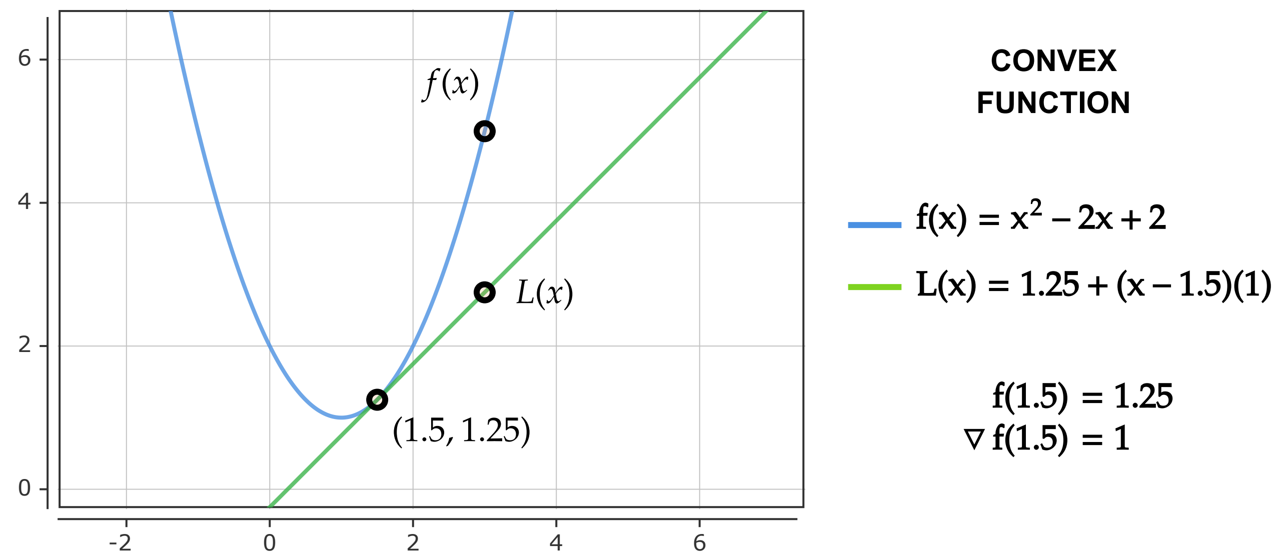
DEFINITION 3

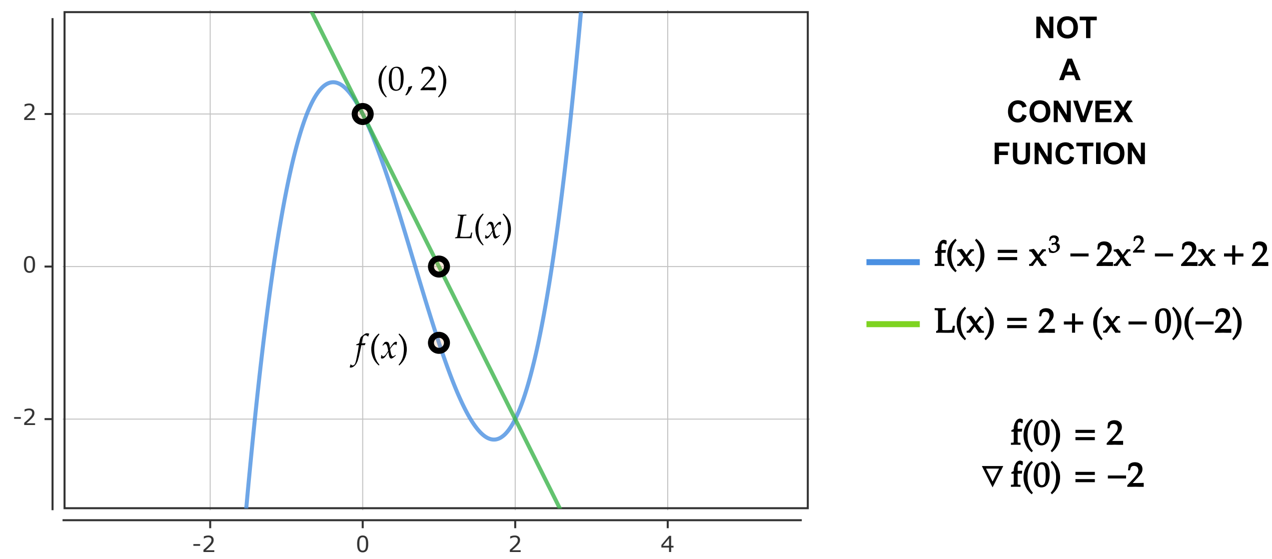
Let $D \subseteq \mathbb{R}^d$ be a convex set. A **differentiable** function $f : D \to \mathbb{R}$ is a convex function if and only if

$$f(x) \geqslant f(a) + (x-a)^T \triangledown f(a)$$

for all $a \in D$

In simple terms, this means that the value of the function at any point is always greater than or equal to its linear approximation at that point.





DEFINITION 4

Let $D \subseteq \mathbb{R}^d$ be a convex set. Let a function $f : D \rightarrow \mathbb{R}$ be twice differentiable. $\mathbf{H}$ is a $d \times d$ matrix such that, for all $i, j \in \{1, 2, \ldots, d\}$

$$H_{ij} = \frac{\delta f^2}{\delta x_i \delta x_j}$$

f is a convex function if and only if $H_f(v)$ is a **positive semi-definite** matrix for all $v \in D$.

If f is a function of one variable, $i.e.,$ if $D \subseteq \mathbb{R}$, then f is convex if

$$f''(x) \geqslant 0, \ \forall x \in D$$

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5. Properties of Convex Functions

PROPERTY 1

The domain of a convex function is always a convex set.

PROOF

Let $f : D \rightarrow \mathbb{R}$ be a convex function. Let us assume that D is not a convex set.

Since D is not a convex function, for some $x, y \in D$, where $x \neq y$, there exists a $\lambda \in [0,1]$ such that

$$\lambda x + (1 - \lambda)y \notin D$$

Since f is a convex function, we can say:

$$f(\lambda x + (1 - \lambda)y) \leqslant \lambda f(x) + (1 - \lambda)f(y)$$

Since f is convex, the above inequality must hold true. But, $\lambda x + (1 - \lambda)y$ is not in D. f cannot be defined at $\lambda x + (1 - \lambda)y$. There is a contradiction. Our assumption is incorrect. Therefore, f is not a convex function.

PROPERTY 2

Let $D \in \mathbb{R}^d$ be a convex set. If $f : D \rightarrow \mathbb{R}$ is a convex function, then all local minima of f are also global minima.

PROOF

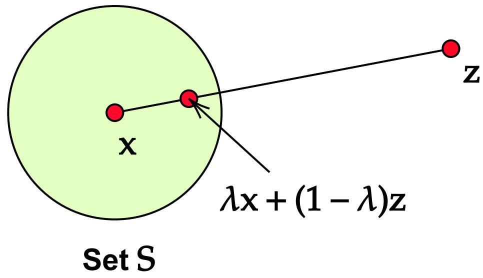
Assume the following

- x is a local minimum of f but **not** a global minimum of f
- z is a global minimum of f
- $f(z) < f(x)$

Since x is a local minimum, we can say that there exist a $\delta > 0$ such that $f(x) \leqslant f(v)$ for all v in the following set.

$$S = \left\{ v \middle| ||x - v|| \leqslant \delta \right\}$$

The above set is represented by the following circle. Let us take a small step from x towards z such that we stay within set S. It is given by $\boldsymbol{\lambda}\mathbf{x} + (\mathbf{1} - \boldsymbol{\lambda})\mathbf{z}$, $\lambda \in [0,1]$.



From our assumption, $f(z) < f(x)$. Therefore,

$$\lambda f(x) + (1-\lambda) f(z) < \lambda f(x) + (1-\lambda) f(x)$$

$\boldsymbol{\lambda}\mathbf{x} + \mathbf{(1-\boldsymbol{\lambda})z} \in S$. Therefore,

$$\begin{aligned} f(x) &\leqslant f(\lambda x + (1-\lambda) z) \\ f(x) &\leqslant \lambda f(x) + (1-\lambda) f(z) \\ f(x) &< \lambda f(x) + (1-\lambda) f(x) \\ f(x) &< f(x) \end{aligned}$$

$f(x) < f(x)$ is a contradiction. Therefore, our assumption that $f(z) < f(x)$ is incorrect. Therefore, x is also a global minimum of f.

PROPERTY 3

Let $D \in \mathbb{R}^d$ be a convex set and let $f: D \to \mathbb{R}$ be a differentiable convex function. $z \in D$ is a global minimum if and only if $\nabla f(z) = 0$.

PROOF

To prove that the above property is true, we must show that the following are true.

- If z is a global minimum, then $\nabla f(z) = 0$
- If $\nabla f(z) = 0$, then f is a global minimum

(A) If z is a global minimum, then $\nabla f(z) = 0$

Let us assume that z is a global minimum of f and $\nabla f(z) \neq 0$. Since $\nabla f(z) \neq 0$, $-\nabla f(z)$ is the direction of steepest descent at z.

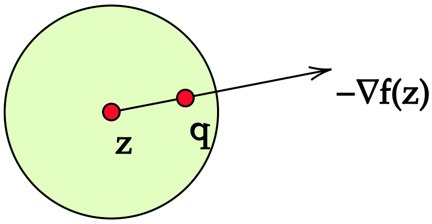
Since z is a global minimum, z is also a local minimum. There exists a $\delta > 0$ such that $f(z) \leqslant f(y)$ for all $y \in S$, where

$$S = \left\{ y \middle| ||z - y|| \leqslant \delta \right\}$$

From z, let us move to q, where

$$q = z - \beta \frac{\nabla f(z)}{||\nabla f(z)||}$$

and $0 < \beta < \delta$.



Set S

Since $-\nabla f(z)$ is the direction of steepest descent,

$$f(q) < f(z)$$

Since $0 < \beta < \delta,\ \ q \in S$. Since $q \in z$, $f(q) \geqslant f(z)$.

We have a contradiction. Our assumption is incorrect. Therefore, $\nabla f(z) = 0$

(B) If $\nabla f(z) = 0$, then f is a global minimum

From the definition of convex function, for all $y \in D$,

$$f(y) \geqslant f(z) + \nabla f(z)^T (y - z)$$

Since $\nabla f(z) = 0$, we get

$$f(y) \geqslant f(z)$$

Therefore, z is a global minimum.

PROPERTY 4

The sum of two convex functions is a convex function.

PROOF

Let $D \in \mathbb{R}^d$ be a convex set and let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ be convex functions. We have to show that $h : D \to \mathbb{R}$ is a convex function, where

$$h(x) = f(x) + g(x),\ \ \forall x \in D$$

Let $u, v \in \mathbb{R}^d$ and let $\lambda \in [0, 1]$

$$\begin{aligned} h(\lambda u + (1-\lambda)v) &= f(\lambda u + (1-\lambda)v) + g(\lambda u + (1-\lambda)v) \\ &\leqslant \lambda f(u) + (1-\lambda)f(v) + \lambda g(u) + (1-\lambda)g(v) \\ &\leqslant \lambda[f(u) + g(u)] + (1-\lambda)[f(v) + g(v)] \\ &\leqslant \lambda h(u) + (1-\lambda)h(v) \end{aligned}$$

Therefore, $h(x)$ is a convex function.

PROPERTY 5

Let D be a convex set. Let $f : \mathbb{R} \to \mathbb{R}$ be a non-decreasing convex function and let $g : D \to \mathbb{R}$ be a convex function. Then, h is a convex function, where

$$h(x) = f(g(x)), \ \forall x \in D$$

PROOF

Since f is a non decreasing function, we can say that for all $x, y \in \mathbb{R}$, if $x \leqslant y$, then

$$f(x) \leqslant f(y)$$

Let $u, v \in D$ and let $\lambda \in [0, 1]$

$$g(\lambda u + (1-\lambda)v) \leqslant \lambda g(u) + (1-\lambda)g(v)$$

Thereofore,

$$\begin{aligned} h(\lambda u + (1-\lambda)v) &= f\big(g(\lambda u + (1-\lambda)v\big) \\ &\leqslant f\big(\lambda g(u) + (1-\lambda)g(v)\big) \\ &\leqslant \lambda f\big(g(u)\big) + (1-\lambda)f\big(g(v)\big) \\ &\leqslant \lambda h(u) + (1-\lambda)h(v) \end{aligned}$$

Therefore, h is a convex function

PROPERTY 6

Let D be a convex set. Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function and let $g : D \to \mathbb{R}$ be a linear function. Then, h is a convex function, where

$$h(x) = f(g(x)), \ \forall x \in D$$

PROOF

Since g is a linear function, the following is true for all $u, v \in D$ and for all $\alpha, \beta \in \mathbb{R}$

$$g(\alpha u + \beta v) = \alpha g(u) + \beta g(v)$$

Let $\lambda \in [0, 1]$

$$\begin{aligned} h(\lambda u + (1-\lambda)v) &= f\big(g(\lambda u + (1-\lambda)v)\big) \\ &= f\big(\lambda g(u) + (1-\lambda)g(v)\big) \\ &\leqslant \lambda f\big(g(u)\big) + (1-\lambda)f\big(g(v)\big) \\ &\leqslant \lambda h(u) + (1-\lambda)h(v) \end{aligned}$$

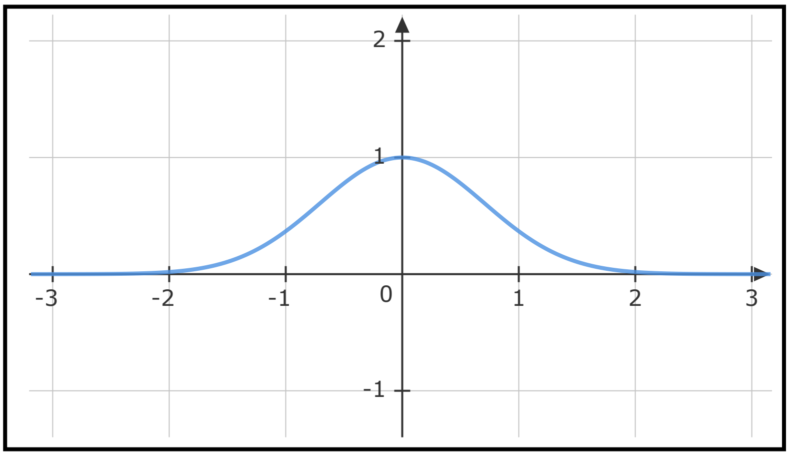
Therefore, h is a convex function

Caveat

In general, the composition of two convex functions need not be convex. Here is an example.

$$f(x) = e^{-x}$$
$$g(x) = x^2$$

$h(x) = e^{-x^2}$ is not a convex function.



$\mathbf{h(x) = e^{-x^2}}$

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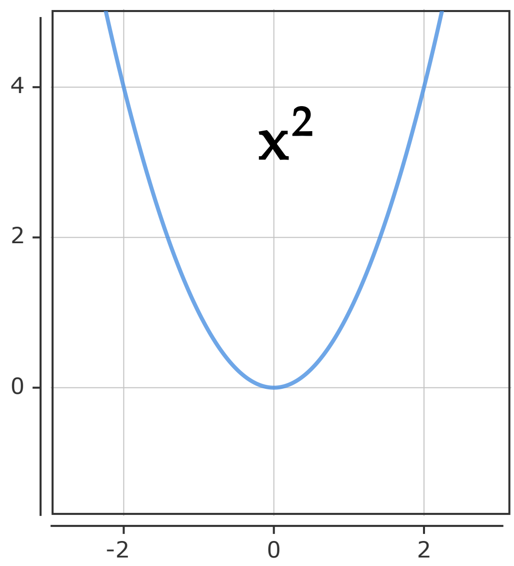
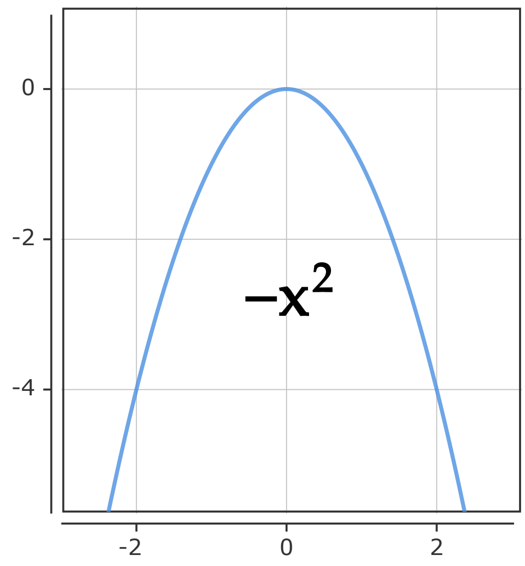
6. Concave Function

A function $f(x)$ is concave if $-f(x)$ is convex.





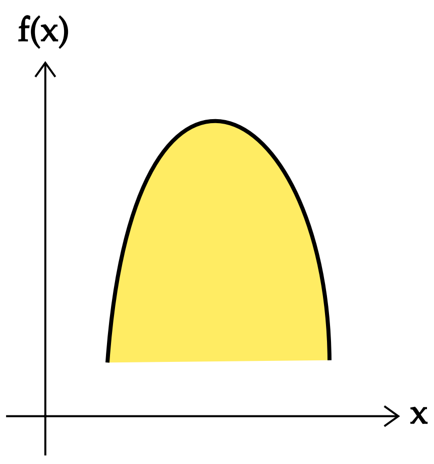
The definitions and properties of concave functions are similar to that of convex functions. Therefore, we will list them without proofs.

DEFINITION 1

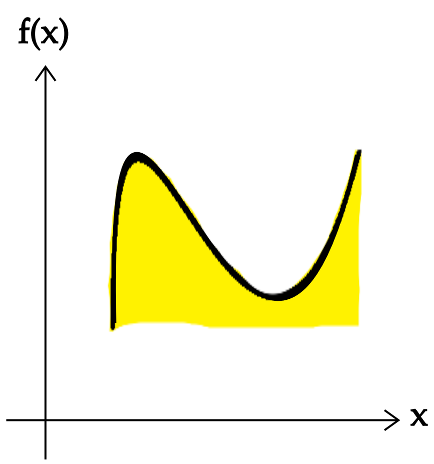
A function $f : D \to \mathbb{R}$ is a concave function if the hypograph of f, denoted by **hyp(f)** is a convex set.

$$\textbf{hyp}(f) = \left\{ \begin{bmatrix} x \\ z \end{bmatrix} \in \mathbb{R}^{d+1} \middle| z \leqslant f(x) \right\}$$

Here, $\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \cdots & x_d & z \end{bmatrix}^T$.



CONVEX SET

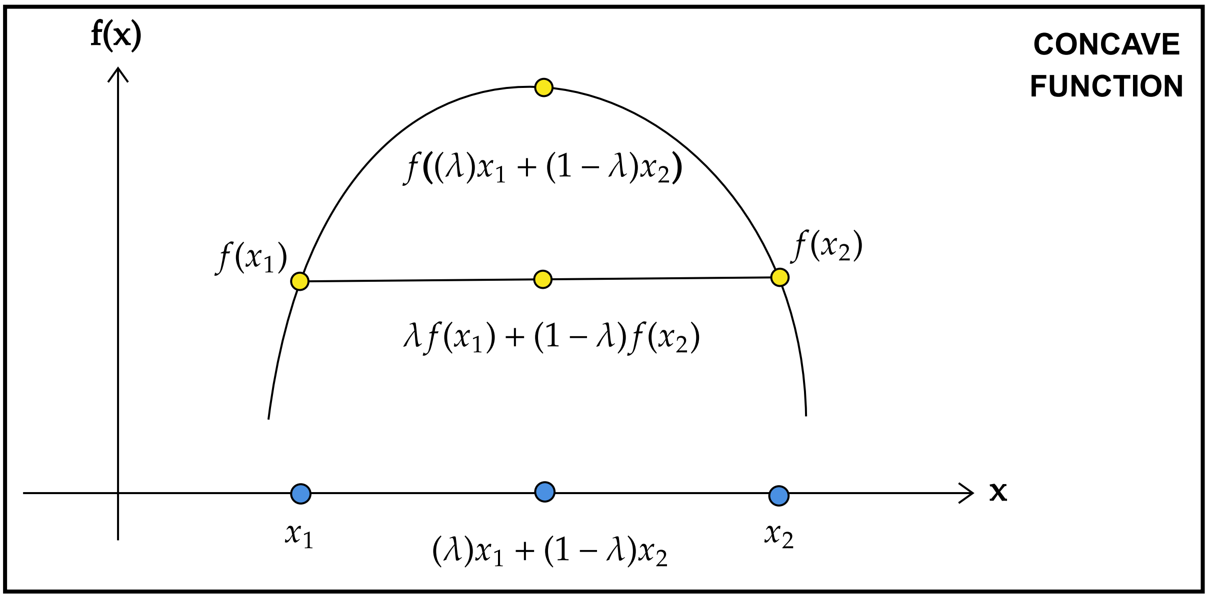


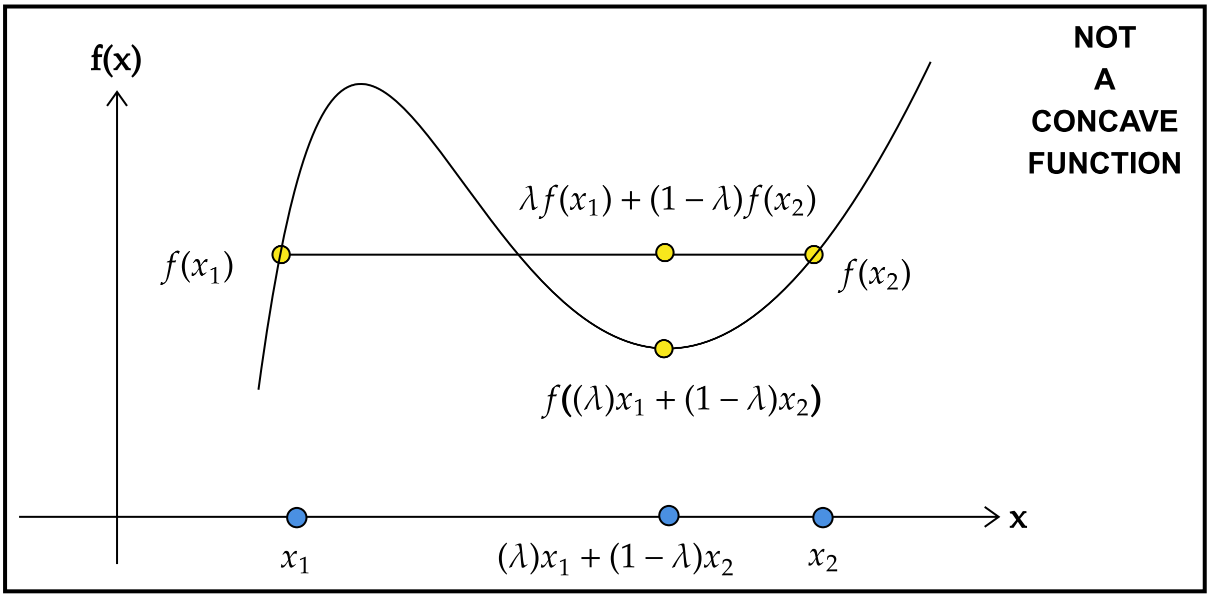
NOT A CONVEX SET

DEFINITION 2

A function $f : D \to \mathbb{R}$ is a concave function if and only if $\forall x_1, x_2 \in \mathbb{R}^d$ and for all $\lambda \in [0, 1]$,

$$f((\lambda)x_1 + (1-\lambda)x_2) \geqslant \lambda f(x_1) + (1-\lambda)f(x_2)$$





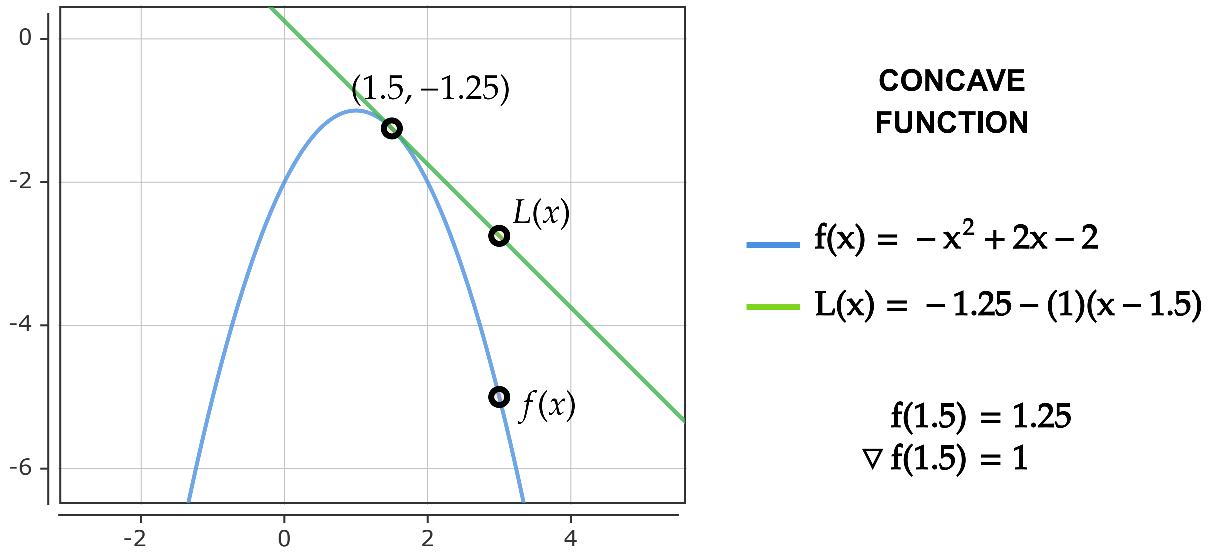
DEFINITION 3

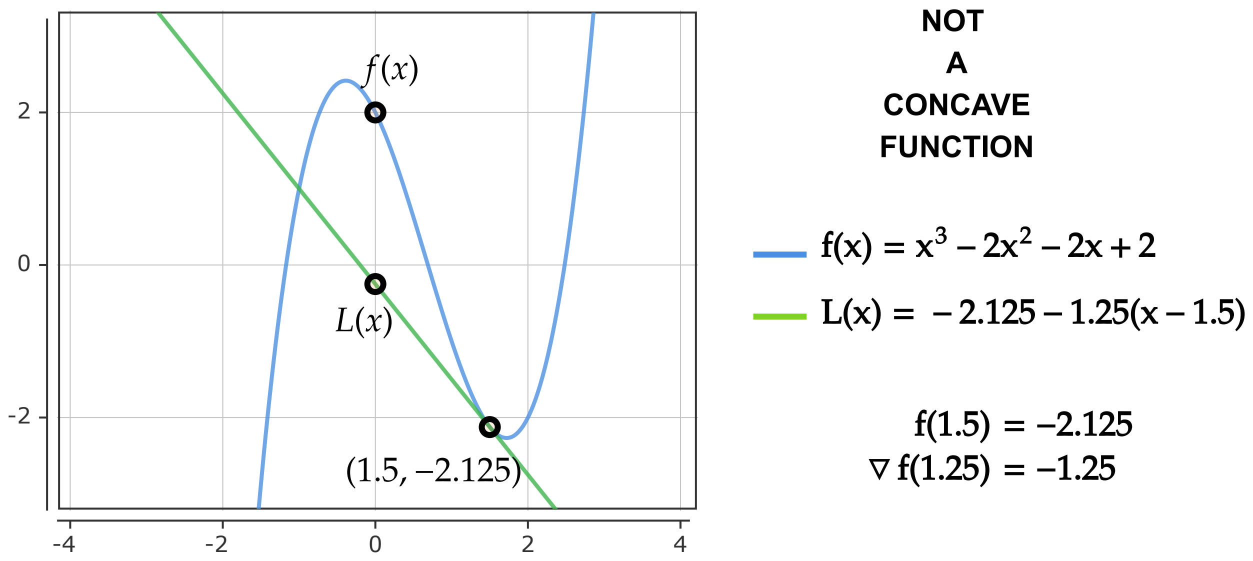
A **differentiable** function $f : D \rightarrow \mathbb{R}$ is a concave function if and only if

$$f(x) \leqslant f(a) + (x-a)^T \nabla f(a)$$

for all $a \in D$

In simple terms, this means that the value of the function at any point is always less than or equal to its linear approximation at that point.





DEFINITION 4

Let a function $f : D \rightarrow \mathbb{R}$ be twice differentiable. $\mathbf{H}$ is a $d \times d$ matrix such that for all $i, j \in \{1, 2, \ldots, d\}$

$$H_{ij} = \frac{\delta f^2}{\delta x_i \delta x_j}$$

f is a concave function if and only if $H_f(v)$ is a **negative semi-definite** matrix for all $v \in D$.

If f is a function of one variable, $i.e.,$ if $D \in \mathbb{R}$, then f is concave if

$$f''(x) \leqslant 0, \;\forall x \in D$$

PROPERTIES

- The domain of a concave function is always a convex set.
- Let f be a concave function. All the local maxima of f are also the global maxima of f.

- Let f be a differentiable concave function. z is a global maximum of f if and only if $\nabla f(z) = 0$.
- The sum of two concave functions is concave.
- Let $D \in \mathbb{R}^d$ be a convex set. Let $f : \mathbb{R} \to \mathbb{R}$ be a non-decreasing concave function and let $g : D \to \mathbb{R}$ be a concave function. Then, $h(x) = f(g(x)),\ \forall x \in D$ is a concave function.
- Let $D \in \mathbb{R}^d$ be a convex set. Let $f : \mathbb{R} \to \mathbb{R}$ be a concave function and let $g : D \to \mathbb{R}$ be a linear function. Then, $h(x) = f(g(x)),\ \forall x \in D$ is a concave function.
- **Caveat :** The composition of two concave function, need not be concave, Let $f(x) = -e^x$ and let $g(x) - x^2$. f and g are concave. However, $h(x) = f(g(x)) = -e^{-x^2}$ is not concave

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7. Necessary vs Sufficient vs Necessary & Sufficient Conditions (Meaning)

Through an example let us understand the meaning of **necessary condition** and **sufficient condition** and **necessary and sufficient condition**

Statement : A number n is divisible by 6

- **Necessary Condition :** n is divisible by 2
- **Sufficient Condition :** n is divisible by 12
- **Necessary & Sufficient Conditon :** n is divisible by 2 and 3

- A **necessary condition** is something that must be true, but by itself does not guarantee the result.
- A **sufficient condition** is something that guarantees the result. but is not required for the statement to be always true.
- A **necessary and sufficent condition** is something that guarantees the result and is required for the statement to be true.

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8. Constrained Optimization

General Form

$$\min_x f(x)$$
$$s.t.$$
$$p_1(x) \leqslant 0, \ldots, p_n(x) \leqslant 0$$
$$q_1(x) = 0, \ldots, q_m(x) = 0$$

- $f(x)$ is the objective function
- $p_1(x) \leqslant 0, \ldots, p_n(x) \leqslant 0$ are the inequality constraints
- $q_1(x) = 0, \ldots, q_m(x) = 0$ are the equality constraints

Lagrangian Function

For the general form of constrained optimization, the following function is called the lagrangian function.

$$L(x, \alpha, \beta) = f(x) + \sum_{i=1}^{n} \alpha_i p_i(x) + \sum_{j=1}^{m} \beta_j q_j(x)$$

where,

$$\alpha = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{pmatrix}$$

$\alpha_1, \ldots, \alpha_n$ and $\beta_1, \ldots, \beta_m$ are called the lagrange multipliers.

The multiplers of inequalty constraints must be non-negative. That is, $\alpha_i \geqslant 0, \ \forall i \in \{1, \ldots, n\}$.

Primal Formulation

Let us focus on

$$\begin{gathered} \min_x f(x) \\ s.t. \\ g(x) \leqslant 0 \end{gathered}$$

The above formulation is referred to as the **primal formulation**.

Consider the Lagrangian function, $(\lambda \geqslant 0)$

$$L(x, \lambda) = f(x) + \lambda g(x)$$

For a fixed $\widehat{x}$ in the domain of f, let us consider the following.

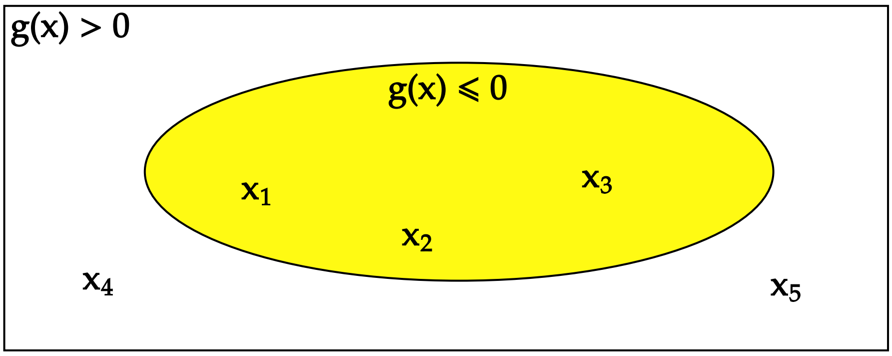
$$\max_{\lambda \geqslant 0} f(\widehat{x}) + \lambda g(\widehat{x})$$

- If $h(\widehat{x}) \leqslant 0$, then $f(\widehat{x}) + \lambda g(\widehat{x})$ is maximum when $\lambda = 0$.
- If $h(\widehat{x}) > 0$, then $\max_{\lambda \geqslant 0} f(\widehat{x}) + \lambda g(\widehat{x}) = 0$. As we increase λ, $\lambda g(\widehat{x})$ increases. Since we are maximizing with respect to $\lambda \geqslant 0$, we can arbitrarily increase λ.

In the figure, given below, let the yellow region be the constraint set S.

$$S = \{x | g(x) \leqslant 0\}$$

Outside the yellow region, the constraint $g(x) \leqslant 0$ is not met.



x_1, x_2 and x_3 are within the constraint set and x_4 and x_5 are outside the constraint set. Let us fix these points and maximize the Lagrangian function with respect to $\lambda \geqslant 0$

Fix x_1

$$\max_{\lambda \geqslant 0} f(x_1) + \lambda g(x_1) = f(x_1)$$

Fix x_2

$$\max_{\lambda \geqslant 0} f(x_2) + \lambda g(x_2) = f(x_2)$$

Fix x_2

$$\max_{\lambda \geqslant 0} f(x_3) + \lambda g(x_3) = f(x_3)$$

Fix x_4

$$\max_{\lambda \geqslant 0} f(x_4) + \lambda g(x_4) = \infty$$

Fix x_5

$$\max_{\lambda \geqslant 0} f(x_5) + \lambda g(x_5) = \infty$$

Suppose, the domain of f is $\{x_1, x_2, x_3, x_4, x_5\}$, then

$$\begin{matrix} \min\limits_x f(x) \\ s.t \\ g(x) \leqslant 0 \end{matrix} = \min \left(f(x_1), f(x_2), f(x_3) \right)$$

This is true as $g(x) \leqslant 0$ is satisfied only at x_1, x_2 and x_3. Similarly,

$$\min_x \max_{\lambda \geqslant 0} \left[f(x) + \lambda g(x) \right] = \min \left(f(x_1), f(x_2), f(x_3) \right)$$

From the above illustration, it is clear that solving the constrained optimization problem

$$\begin{gathered}\min_x f(x) \\ s.t \\ g(x) \leqslant 0\end{gathered}$$

is equivalent to solving

$$\min_x \max_{\lambda \geqslant 0} \left[f(x) + \lambda g(x) \right]$$

Dual Formulation

We know that the following is true

$$\begin{gathered}\min_x f(x) \\ s.t \\ g(x) \leqslant 0\end{gathered} \equiv \min_x \max_{\lambda \geqslant 0} \left[f(x) + \lambda g(x) \right]$$

The **dual formulation** for the above is given below.

$$\max_{\lambda \geqslant 0} \min_x \left[f(x) + \lambda g(x) \right]$$

In the dual formulation, we have interchanged $\min$ and $\max$. This leads us to two concepts.

- Weak Duality
- Strong Duality

Weak Duality

Consider the following function.

$$J(x) = \begin{cases} f(x), & if\ g(x) \leqslant 0 \\ \infty, & if\ g(x) > 0 \end{cases}$$

For any fixed $\lambda \geqslant 0$

- If $g(x) \leqslant 0$, then $\lambda g(x) \leqslant 0$. Therefore,

$$L(x, \lambda) = f(x) + \lambda g(x) \leqslant f(x)$$

- If $g(x) \leqslant 0$, then $\lambda g(x) \geqslant 0$. Therefore,

$$L(x, \lambda) = f(x) + \lambda g(x) \leqslant \infty$$

Therefore, for any fixed $\lambda \geqslant 0$, and for all x in the domain of f,

$$L(x, \lambda) = f(x) + \lambda g(x) \leqslant J(x) \quad (1)$$

Let x^* be the solution for the primal formulation. That is,

$$x^* = \arg\min_x f(x),\ s.t.,\ g(x) \leqslant 0$$

Note that x^* satisfies $g\left(x^*\right) \leqslant 0$. Therefore,

$$x^* = \arg\min_x J(x)$$

Also,

$$\min_x J(x) = f\left(x^*\right)$$

Equation (1) is true for any fixed $\lambda \geqslant 0$ for all x in the domain of f. Therefore,

$$\min_x \left[f(x) + \lambda g(x)\right] \leqslant \min_x J(x)$$
$$\min_x \left[f(x) + \lambda g(x)\right] \leqslant f\left(x^*\right)$$

The above inequality is true for any $\lambda \geqslant 0$. Therefore,

$$\max_{\lambda \geqslant 0} \min_x \left[f(x) + \lambda g(x)\right] \leqslant f\left(x^*\right)$$

The left hand side of the inequality is the dual formulation. The right hand side is the value of $f(x)$ at the primal optimal x^*. The dual formulation provides a **lower bound** for the **optimal** value of primal formulation. This property is called the **weak duality**.

Strong Duality

If $f(x)$ and $g(x)$ are convex, and they satisfy some regularity conditions (for our purpose, we can ignore those), then the following is true.

$$\max_{\lambda \geqslant 0} \min_x \left[f(x) + \lambda g(x)\right] = f\left(x^*\right)$$

This property is referred to as the **strong duality**.

When strong duality holds, we can either solve the primal problem or the dual problem.

KKT Condition (Informal)

Let us define x^* and λ^* as follows,

$$x^* = \arg\min_x f(x),\ s.t.,\ g(x) \leqslant 0$$

$$\lambda^* = \arg\max_{\lambda \geqslant 0} \min_x \left[f(x) + \lambda g(x) \right]$$

x^* is the primal optimal and λ^* is the dual optimal.

Suppose, strong duality holds, then what are the **necessary** conditions that x^* and λ^* satisfy?

Stationarity Condition and Complementary Slackeness Condition

Let us focus on the Lagrangian function at x^* and λ^*

$$L\left(x^*, \lambda^*\right) = f\left(x^*\right) + \lambda^* g\left(x^*\right)$$

$\lambda^* g\left(x^*\right) \leqslant 0$. Therefore,

$$L\left(x^*, \lambda^*\right) \leqslant f\left(x^*\right)$$

$$f\left(x^*\right) + \lambda^* g\left(x^*\right) \leqslant f\left(x^*\right) \qquad \textbf{(A)}$$

Let us now focus the dual formulation. Since λ^* is the dual optimal,

$$\max_{\lambda \geqslant 0} \min_x \left[f(x) + \lambda g(x) \right] = \min_x \left[f(x) + \lambda^* g(x) \right]$$

We are minimizing $L\left(x, \lambda^*\right) = f(x) + \lambda^* g(x)$ with respect to x. Suppose x' is the argument that minimizes $L\left(x, \lambda^*\right)$

$$x' = \arg\min_x f(x) + \lambda^* g(x)$$

then, $L\left(x', \lambda^*\right) \leqslant L\left(x, \lambda^*\right)$ for all x in the domain of f. Therefore, it should be true for x^*,

$$f(x') + \lambda^* g(x') \leqslant f\left(x^*\right) + \lambda^* g\left(x^*\right) \qquad \textbf{(B)}$$

From inequalities $\textbf{(A)}$ and $\textbf{(B)}$, we get the following.

$$f(x') + \lambda^* g(x') \leqslant f\left(x^*\right) + \lambda^* g\left(x^*\right) \leqslant f\left(x^*\right)$$

We know that

$$f(x') + \lambda^* g(x') = \min_x f(x) + \lambda^* g(x)$$

Therefore,

$$\min_x f(x) + \lambda^* g(x) \leqslant f\left(x^*\right) + \lambda^* g\left(x^*\right) \leqslant f\left(x^*\right)$$

From strong duality, we know that

$$\min_x f(x) + \lambda^* g(x) = f\left(x^*\right)$$

Therefore,

$$\begin{aligned} \min_x f(x) + \lambda^* g(x) = f\left(x^*\right) + \lambda^* g\left(x^*\right) = f\left(x^*\right) \\ f(x') + \lambda^* g(x') = f\left(x^*\right) + \lambda^* g\left(x^*\right) = f\left(x^*\right) \end{aligned}$$

First, let us focus on $f(x') + \lambda^* g(x') = f\left(x^*\right) + \lambda^* g\left(x^*\right)$. Recall

$$x' = \arg\min_x f(x) + \lambda^* g(x)$$

x' is a solution to an unconstrained optimization problem. Therefore, $\nabla f(x') + \lambda^* \nabla g(x') = 0$.

$$\begin{aligned} f(x') + \lambda^* g(x') &= f\left(x^*\right) + \lambda^* g\left(x^*\right) \\ \nabla f(x') + \lambda^* \nabla g(x') &= \nabla f\left(x^*\right) + \lambda^* \nabla g\left(x^*\right) \\ 0 &= \nabla f\left(x^*\right) + \lambda^* \nabla g\left(x^*\right) \end{aligned}$$

$\nabla f\left(x^*\right) + \lambda^* \nabla g\left(x^*\right) = 0$ is called the **stationarity condition**. The gradient of the lagrangian at x^* and λ^* should be equal to 0.

Let us focus on $f\left(x^*\right) + \lambda^* g\left(x^*\right) = f\left(x^*\right)$

$$\begin{aligned} f\left(x^*\right) + \lambda^* g\left(x^*\right) &= f\left(x^*\right) \\ \lambda^* g\left(x^*\right) &= 0 \end{aligned}$$

This is called the **complementary slackness condition**. Either $g\left(x^*\right) = 0$ or $\lambda^* = 0$.

KKT Condition (Formal)

$$\begin{gathered} \min_x f(x) \\ s.t. \\ p_1(x) \leqslant 0, \ldots, p_n(x) \leqslant 0 \\ q_1(x) = 0, \ldots, q_m(x) = 0 \end{gathered}$$

The following is the lagrangian function for the above formulation.

$$L(x, \alpha, \beta) = f(x) + \sum_{i=1}^{n} \alpha_i p_i(x) + \sum_{j=1}^{m} \beta_j q_j(x)$$

where,

$$\alpha = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \ \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{pmatrix}$$

Let us define x^*, α^* and β^* as follows

$$x^* = \arg\min_x \left(\max_{\alpha \geqslant 0,\, \beta} L(x, \alpha, \beta) \right)$$

$$\alpha^*, \beta^* = \arg\max_{\alpha \geqslant 0,\, \beta} \left(\min_x L(x, \alpha, \beta) \right)$$

Suppose, strong duality holds true. Then, the following are **necessary** condition that x^*, α^* and β^* have to satisfy. They are called the **KKT** conditions.

(A) Stationarity Condition

$$\nabla f\left(x^*\right) + \sum_{i=1}^{n} \left(\alpha^*\right) \nabla p_i\left(x^*\right) + \sum_{j=1}^{m} \left(\beta_j^*\right) \nabla q_j\left(x^*\right) = 0$$

(B) Complementary Slackness Condition

$$\alpha_i^* p_i\left(x^*\right) = 0, \ \forall i \in \{1, \ldots, n\}$$

Since $q_j\left(x^*\right) = 0$, is true $\forall j \in \{1, \ldots, m\}$, the condition $\beta_i q_j\left(x^*\right) = 0$ automatically holds true $\forall \beta_j \in \mathbb{R}$. Therefore, it does not impose any additional constraint and is not included as a KKT condition.

(C) Primal Feasibility

$$p_i\left(x^*\right) \leqslant 0, \ \forall i \in \{1, \ldots, n\}$$
$$q_j\left(x^*\right) = 0, \ \forall j \in \{1, \ldots, m\}$$

(D) Dual Feasibility

$$\alpha_i \geqslant 0, \ \forall i \in \{1, \ldots, n\}$$

β_j can be any real number, $\forall j \in \{1, \ldots, m\}$.

Sufficient Condition

Suppose, strong duality holds true. If x^*, α^* and β^* satisfy the **KKT** conditions,

- x^* is the primal optimum
- α^* and β^* are the dual optimal

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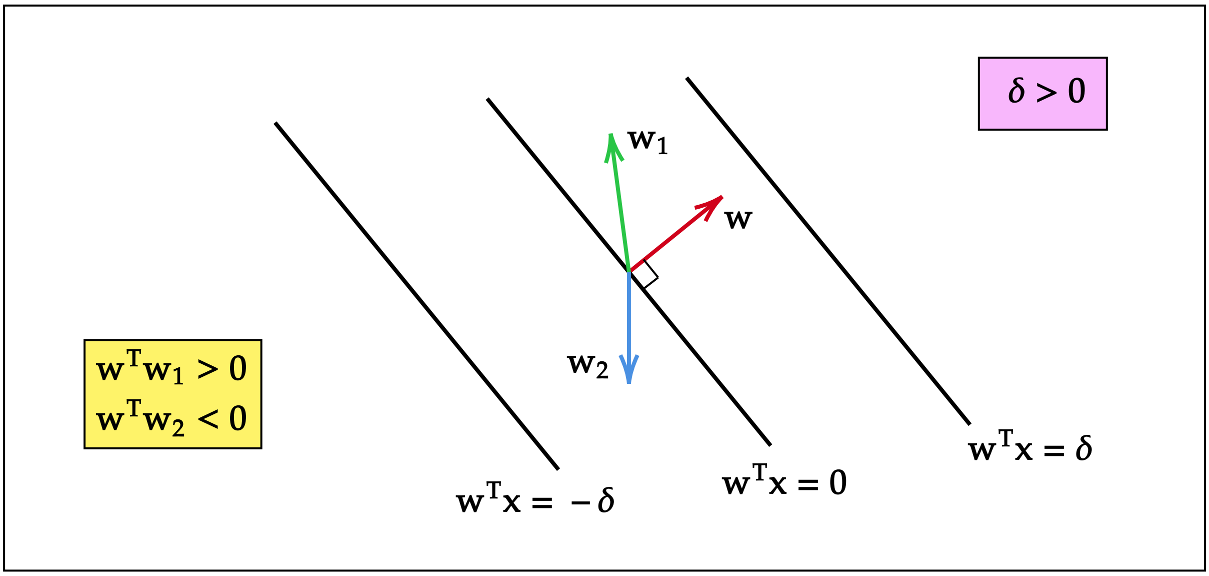
9. Dot Product



- The angle between $\mathbf{w}$ and $\mathbf{w}_1$ is less than $90°$. Therefore, $\mathbf{w}^T\mathbf{w}_1 > 0$.
- The angle between $\mathbf{w}$ and $\mathbf{w}_2$ is greater than than $90°$. Therefore, $\mathbf{w}^T\mathbf{w}_2 < 0$.

---

10. Geometric Perspective of Stationarity Condition

For this section, let us assume that strong duality holds.

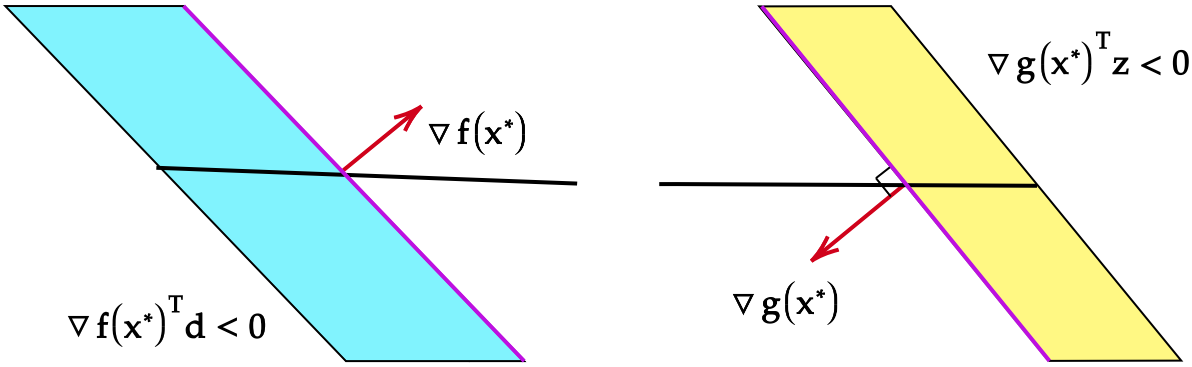
Inequality Constraint

$$\begin{gathered}\min_x f(x) \\ s.t. \\ g(x) \leqslant 0\end{gathered}$$

Let x^* be the primal optimum and let λ^* be the dual optimum. By stationarity condition,

$$\begin{aligned}\nabla f\left(x^*\right) + \lambda^* \nabla g\left(x^*\right) &= 0 \\ \nabla f\left(x^*\right) &= -\lambda^* \nabla g\left(x^*\right)\end{aligned}$$

Since $\lambda^* \geqslant 0,$ the gradient of f at x^* and the gradient of g at x^* are **anti-parallel** to each other.



$$\boxed{\nabla f(x^*) = -(\lambda^*)\, \nabla g(x^*)}$$

From Taylor series,

$$f(x^* + d) \approx f(x^*) + \nabla f(x^*)^T (d)$$
$$f(x^* + d) - f(x^*) \approx \nabla f(x^*)^T (d)$$

and

$$g(x^* + z) \approx g(x^*) + \nabla g(x^*)^T (z)$$
$$g(x^* + z) - g(x^*) \approx \nabla g(x^*)^T (z)$$

The blue region is denoted by

$$\left\{ d \,\middle|\, \nabla f(x^*)^T d \leqslant 0 \right\}$$

Therefore,

$$f(x^* + d) - f(x^*) \leqslant 0$$

Yellow region is denoted by

$$\left\{ z \,\middle|\, \nabla g(x^*)^T z \leqslant 0 \right\}$$

Therefore,

$$g(x^* + z) - g(x^*) \leqslant 0$$

- d is a **descent** direction of f at x^*
- z is a **descent** direction of g at x^*.

Descent Direction : For a function $h(x)$, d is a descent direction at x^*, if for some sufficiently small $\beta > 0$, $h(x + \beta d) < h(x)$.

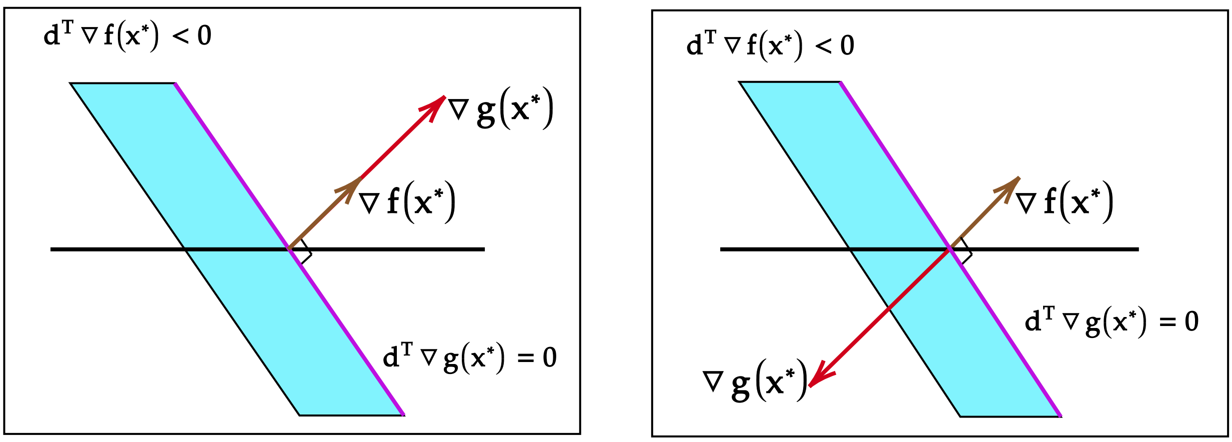
Equality Constraint

$$\min_x f(x)$$
$$s.t.$$
$$g(x) = 0$$

Let x^* be the primal optimum and let λ^* be the dual optimum. By stationarity condition,

$$\begin{aligned} \nabla f(x^*) + \lambda^* \nabla g(x^*) &= 0 \\ \nabla f(x^*) &= -\lambda^* \nabla g(x^*) \end{aligned}$$

For equality constrian, λ can be any real number. Therefore, $\nabla f(x^*)$ and $\nabla g(x^*)$ are scalar multiples of each other.



The pink line represents all the directions d such that :

$$d^T \nabla g(x^*) = 0$$

Along these directions, the value of the function $g(x)$ remains equal to $g(x^*)$. Since x^* is the primal optimal, it satisfies $g(x^*) = 0$. As a result, the value of the function $g(x)$ along the pink line is also 0. For sufficiently small steps along the pink line, the constraint $g(x) = 0$ remains satisfied.

The blue region corresponds to $\left\{ d \middle| d^T \nabla f(x^*) \leqslant 0 \right\}$. The blue region are the descent directions of f at x^*.

Notice that the blue region and the pink line do not intesect. If they were to intersect, then there would exist a direction along the pink line that is also a descent direction for f at x^*. In that case, starting from x^* we could take a sufficiently small step along the pink direction and reduce the zvalue of the objective function $f(x)$ while still satisfying the constraint $g(x) = 0$. This would imply that x^* is not an optimal solution.

---

11. Method Of Lagrange Multipliers (Example)

$$\min_{x_1,x_2} \left[\mathbf{f(x_1,x_2) = x_1^2 + 2x_2 + 4x_2^2}\right]$$

$$s.t\ \mathbf{g(x_1,x_2) =}\ 0$$

$$g(x_1,x_2) = x_1^2\ + x_2^2 - 1$$

Find gradient

$$\nabla f(x_1,x_2) = (2x_1, 2 + 8x_2)$$
$$\nabla g(x_1,x_2) = (2x_1, 2x_2)$$

We have an equaity constraint.

By stationarity condition, $(\lambda \in \mathbb{R})$

$$\nabla f(x_1,x_2) = -\lambda \times \nabla g(x_1,x_2)$$
$$(2x_1, 2 + 8x_2) = -\lambda(2x_1, 2x_2)$$

We get the following equations

$$2x_1 = -\lambda(2x_1) \quad \text{(4)}$$
$$2 + 8x_2 = -\lambda(2x_2) \quad \text{(5)}$$

For (1) to be satisfied, either $\lambda = -1$ or $x_1 = 0$.

CASE 1 : $-\lambda = 1$

Substitute $-\lambda = 1$ in (2)

$$2 + 8x_2 = 2x_2$$
$$6x_2 = -2$$

$$x_2 = (-1/3)$$

By primal feasibility, $g(x_1, x_2) = 0$. Substitute $x_2 = (-1/3)$ in $g(x_1, x_2) = 0$

$$\begin{aligned} x_1^2 + (-1/3)^2 - 1 &= 0 \\ x_1^2 &= (8/9) \\ x_1 &= \pm\sqrt{8/9} \end{aligned}$$

CASE 2 : $\mathbf{x_1 = 0}$

Substitute $x_1 = 0$ in $g(x_1, x_2) = 0$

$$\begin{aligned} 0^2 + x_2^2 - 1 &= 0 \\ x_2 &= \pm 1 \end{aligned}$$

Possible solutions

$$\left\{ \begin{pmatrix} \sqrt{8/9} \\ -1/3 \end{pmatrix}, \begin{pmatrix} -\sqrt{8/9} \\ -1/3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\}$$

Find the value of f at these points.

$$f\begin{pmatrix} \sqrt{8/9} \\ -1/3 \end{pmatrix} = \frac{2}{3} \qquad f\begin{pmatrix} -\sqrt{8/9} \\ -1/3 \end{pmatrix} = \frac{2}{3} \qquad f\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 6 \qquad f\begin{pmatrix} 0 \\ -1 \end{pmatrix} = 2$$

Therefore, $\begin{pmatrix} \sqrt{8/9} \\ -1/3 \end{pmatrix}$ and $\begin{pmatrix} -\sqrt{8/9} \\ -1/3 \end{pmatrix}$ are the optimal solutions.

---

12. Projected Gradient Descent

It might be difficult to solve constrained optimization analytically, using the method of Lagrange multipliers. Here is an iterative algorithm.

GOAL

$$\min_x [f(x)]$$

$$s.t\ \mathbf{g(x) \leqslant 0}$$

Let $\eta_t > 0$

STEP 1 : Initialize x_0

STEP 2 : For **t** in $\{0, 1, 2, ..., T\}$

$$x_{t+1} = \prod \left(x_t - \eta \triangledown f(x_t)\right)$$

STEP 3 : End

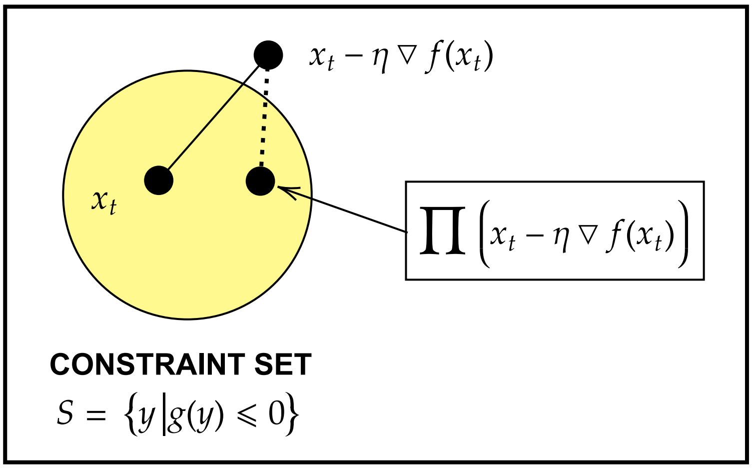
- **Gradient Step :** $x_t - \eta \triangledown f(x_t)$
- **Projection Step :** $\prod \left(x_t - \eta \triangledown f(x_t)\right)$

Here,

$$\prod (x) = \arg\min_{y \in S} \|x - y\|^2$$

where

$$S = \{y | g(y) \leqslant 0\}$$



In the above picture, the projection step finds the point set S, that is closest to $x_t - \eta \triangledown f(x_t)$.

If the constraint set is **convex**, then projected gradient descent converges to an optimal solution, under suitable conditions on the objective function.

---

13. Stochastic Gradient Descent

Let us revisit linear regression $(\text{week} - 4)$. Consider the following dataset.

$$D = \{(x_1, y_1), ..., (x_n, y_n)\}$$

where, $x_i \in \mathbb{R}^d$, and $y \in \mathbb{R}$, $\forall i \in \{1, ..., n\}$

- $x_i \rightarrow$ data point
- $y_i \rightarrow$ corresponding label

Consider matrix X and Y

$$X = \begin{bmatrix} - & x_1 & - & 1 \\ - & x_2 & - & 1 \\ & \vdots & & \vdots \\ - & x_n & - & 1 \end{bmatrix}_{\mathbf{n \times (d+1)}} \quad , \; Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{\mathbf{n \times 1}}$$

Every row of X is a data point. All the entries in the last column of X are 1. The goal is to find $\widehat{\mathbf{w}} \in \mathbb{R}^{d+1}$ such that

$$\widehat{\mathbf{w}} = \arg\min_{\mathbf{w}} \left|\left|X\mathbf{w} - Y\right|\right|^2$$

Let $f(\mathbf{w}) = \left|\left|X\mathbf{w} - Y\right|\right|^2$. $f(\mathbf{w})$ can also be written as follows :

$$f(\mathbf{w}) = \sum_{i=1}^{n} \left(\mathbf{w}^T x_i - y_i\right)^2$$

Let us focus on $h_i(\mathbf{w}) = \left(\mathbf{w}^T x_i - y_i\right)^2$. $h_i(\mathbf{w})$ can be written as follows :

$$h_i(\mathbf{w}) = p(q_i(\mathbf{w}))$$

where,

$$\begin{aligned} p(x) &= x^2 \\ q_i(\mathbf{w}) &= \mathbf{w}^T x_i + y_i \end{aligned}$$

$q_i(\mathbf{w})$ is a linear function, and $p(x)$ is a convex function. The compositon of a convex function and a linear function is convex. Therefore, $h_i(\mathbf{w})$ is convex.

$f(\mathbf{w})$ is a sum of convex functions, and is therefore, convex.

- Every local minima of f is also a global minima
- $\widehat{\mathbf{w}}$ is a global minima if and only if $\nabla f(\widehat{\mathbf{w}}) = 0$

$$\widehat{\mathbf{w}} = \arg\min_{\mathbf{w}} \sum_{i=1}^{n} \left(\mathbf{w}^T x_i - y_i\right)^2$$

The gradient of f is given as follows :

$$\nabla f(\mathbf{w}) = X^T X\mathbf{w} - X^T Y$$

Setting $\nabla f(\mathbf{w}) = 0$

$$X^TX\mathbf{w} - X^TY = 0$$
$$\mathbf{w} = \left(X^TX\right)^+ X^TY$$

$+$ symbol denotes the **psuedoinverse.** If X^TX has an inverse, then

$$\mathbf{w} = \left(X^TX\right)^{-1} X^TY$$

The time complexity of computing $\left(X^TX\right)^{-1}$ is $O\left(d^3\right)$. If d is large, then computing $\left(X^TX\right)^{-1}$ would be computationally costly.

For the objective function, every local minimum is also a global minimum. We know that the gradient descent algorithm converges to a local minimum. If d is large, instead of computing $\left(X^TX\right)^{-1}$, we can use **gradient descent** find $\widehat{\mathbf{w}}$. Here is the algorithm. Let $\eta_t = \dfrac{1}{t+1}$

- Initialize $\mathbf{w_0}$
- for $t \in \{0,..., T\}$
  - $\mathbf{w_{t+1}} = \mathbf{w_t} - \eta_t\left(X^TX\mathbf{w_t} - X^TY\right)$
- End

To compute the gradient, we need not find the inverse of X^TX.

X^T is a $d \times n$ matrix and X is an $n \times d$ matrix. Suppose, n, which corresponds to the number of data points is large, then computing X^TX would be costly. In such cases, we can make use of an algorithm called the **stochastic gradient descent**. Here is the algorithm. Let $\eta_t = \dfrac{1}{t+1}$

- Initialize $\mathbf{w_0}$
- For $t \in \{0,..., T-1\}$
  - Sample k data points $\left(\text{where } k \ll n\right)$, and their corresponding lables from $\{(x_1, y_1),..., (x_n, y_n\})$.
  - Let the sampled data points be $p_1,..., p_k$ and let their corrsponding lables be $q_1,..., q_k$
  - Consider the matrices P and Q

$$P = \begin{bmatrix} — & p_1 & — & 1 \\ — & p_2 & — & 1 \\ & \vdots & & \vdots \\ — & p_k & — & 1 \end{bmatrix}_{\mathbf{k \times (d+1)}}, \; Q = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix}_{\mathbf{k \times 1}}$$

- Update as follows

$$\mathbf{w_{t+1}} = \mathbf{w_t} - \eta_t \left(P^T P \mathbf{w_t} - P^T Q \right)$$

- End

It can be shown that the average of

$$\frac{1}{T+1} \left(\sum_{t=0}^{T} \mathbf{w}_t \right) \rightarrow \widehat{\mathbf{w}}$$
